

University of Information Technology & Sciences (UITs)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Mid Term Examination, Autumn 2025
Course Title: Electronic Device and Circuits
Course Code: EEE 0714123

Marks: 20

Time: 1(One) hour

(Answer all questions)

1. a) Explain the effect of forward and reverse bias on the depletion region and relate it to the diode acting as a one-way valve for current. [05]
- b) Analyze the circuit of Figure 1. to find the voltage across the load and the currents through the load, Zener diode, and series resistor for $R_L=180\ \Omega$. [05]

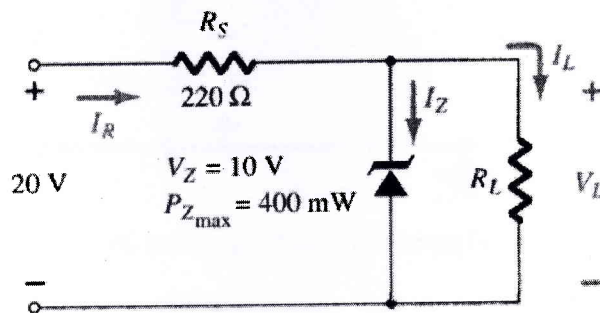


Figure 1 : Circuit for 1b

2. a) For the rectifier circuit in Figure 2, Draw the output voltage waveform V_O and calculate the resulting DC voltage. [05]

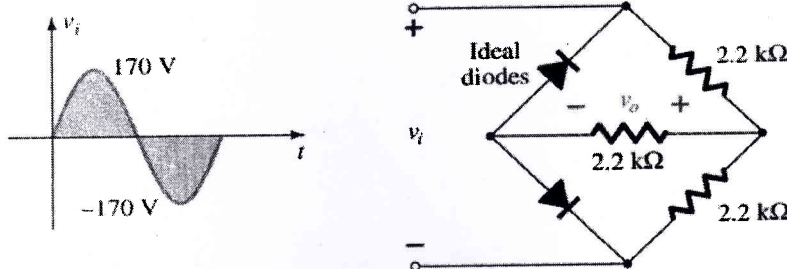


Figure 2: Circuit for question 2a

2 questions

- b) **Design** a series positive clipper using a diode and resistor. **Sketch** the output wave shape with necessary explanation for the circuit shown in Figure 3. [05]

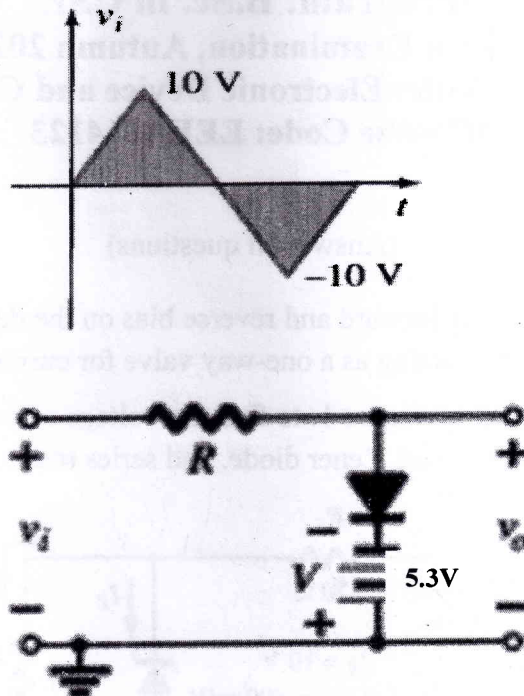


Figure 3: Circuit for question 2b